

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	58386	52.133150
1	[Git, Change, Create]	17242	15.395468
2	[Git, Commit]	11257	10.051431
3	[Git, Change, Delete]	8758	7.820062
4	[File, Git]	5891	5.260103
5	[Git, Change, Rename]	3354	2.994803
6	[Git, Tag]	1732	1.546511
7	[Author, Git, Person]	1265	1.129525
8	[Json, Key]	668	0.596461
9	[Json, Value, Scalar]	603	0.538422
10	[Committer, Git, Person]	370	0.330375
11	[NPM, Dependency]	338	0.301802
12	[Type, TS, Primitive]	293	0.261621
13	[Type, TS, Declared]	277	0.247335
14	[TS, ExternalDeclaration]	215	0.191975
15	[Git, Change, Copy]	138	0.123221
16	[Type, TS, Literal]	136	0.121435
17	[Json, Value, Object]	133	0.118756
18	[Type, TS, Union]	120	0.107149
19	[Type, TS, ObjectMember]	103	0.091969
20	[NPM, Script]	91	0.081254
21	[TS, Property]	65	0.058039
22	[TS, Function]	47	0.041967
23	[Git, Branch]	43	0.038395
24	[Type, TS, FunctionParameter]	40	0.035716
25	[Type, Object, TS]	39	0.034823
26	[File, Directory]	34	0.030359
27	[Type, TS, Function]	34	0.030359
28	[TS, Parameter]	33	0.029466
29	[Package, File, Json, NPM]	29	0.025894

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

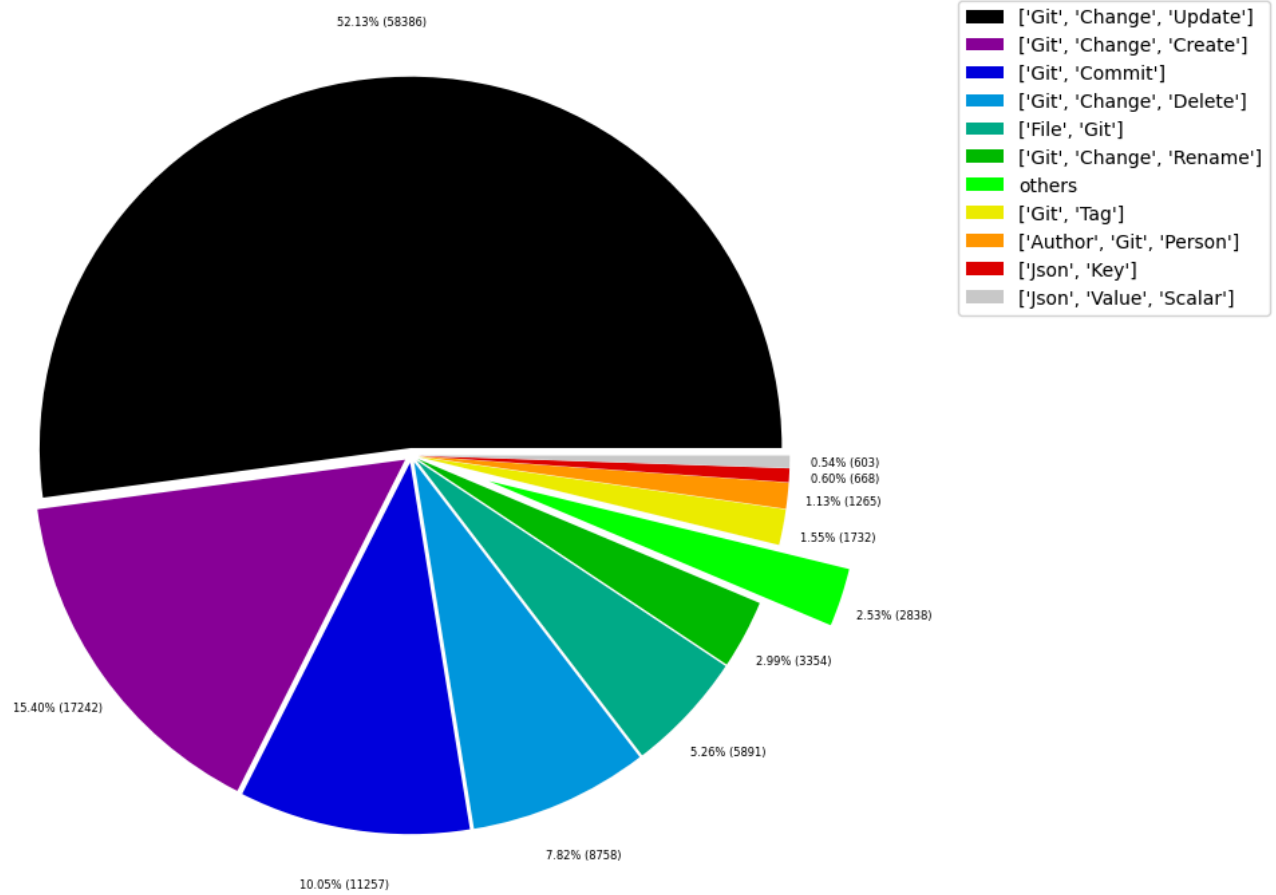


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000893
1	[NPM, PublishConfig]	1	0.000893
2	[File, TS, Scan]	1	0.000893
3	[TS, Method]	1	0.000893
4	[Repository, File, Git]	1	0.000893
5	[TS, Constructor]	1	0.000893
6	[Value, TS, ObjectMember]	1	0.000893
7	[TS, Class]	1	0.000893
8	[TS, Enum]	2	0.001786
9	[Value, Object, TS]	3	0.002679
10	[Type, TS, Tuple]	3	0.002679
11	[Value, TS, Function]	4	0.003572
12	[TS, TypeParameter]	4	0.003572
13	[Value, TS, Complex]	5	0.004465
14	[Repository, NPM]	5	0.004465
15	[NPM, Engine]	6	0.005357
16	[Project, TS]	6	0.005357
17	[File, Local]	6	0.005357
18	[Type, TS, TypeParameterReference]	6	0.005357
19	[Value, TS, Member]	6	0.005357
20	[File, TS, Local, Module]	6	0.005357
21	[Value, TS, Call]	6	0.005357
22	[TS, EnumMember]	8	0.007143
23	[Type, TS, NotIdentified]	11	0.009822
24	[TS, ExternalModule]	11	0.009822
25	[Json, Value, Array]	12	0.010715
26	[Value, TS, Declared]	13	0.011608
27	[TS, TypeAlias]	16	0.014286
28	[File, Directory, Local]	16	0.014286
29	[TS, Interface]	17	0.015179

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

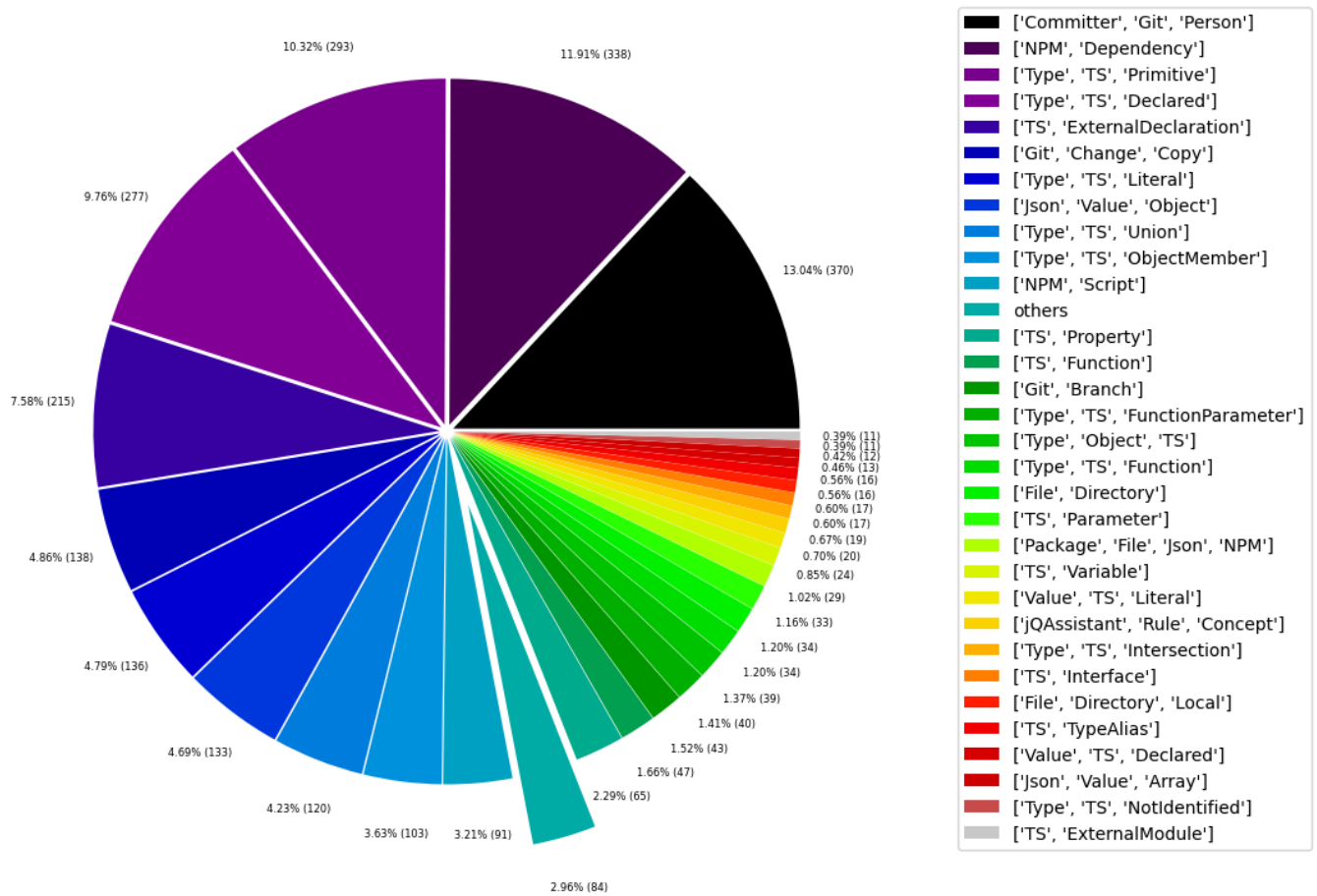


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

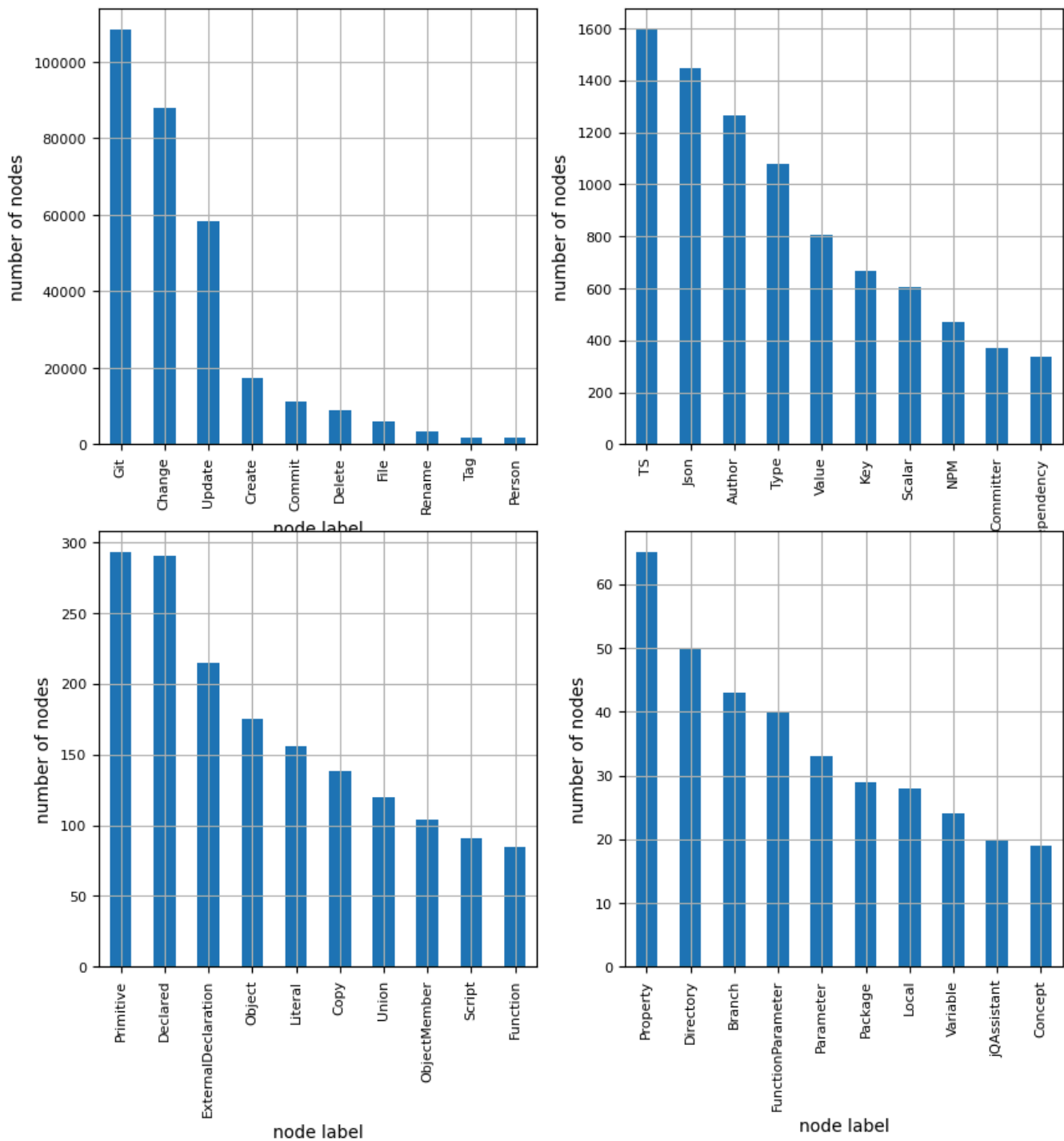
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	108437	96.823937
1	Change	87878	78.466704
2	Update	58386	52.133150
3	Create	17242	15.395468
4	Commit	11257	10.051431
5	Delete	8758	7.820062
6	File	5984	5.343143
7	Rename	3354	2.994803
8	Tag	1732	1.546511
9	Person	1635	1.459900
10	TS	1595	1.424183
11	Json	1445	1.290248
12	Author	1265	1.129525
13	Type	1079	0.963444
14	Value	806	0.719681
15	Key	668	0.596461
16	Scalar	603	0.538422
17	NPM	470	0.419665
18	Committer	370	0.330375
19	Dependency	338	0.301802
20	Primitive	293	0.261621
21	Declared	290	0.258942
22	ExternalDeclaration	215	0.191975
23	Object	175	0.156258
24	Literal	156	0.139293
25	Copy	138	0.123221
26	Union	120	0.107149
27	ObjectMember	104	0.092862
28	Script	91	0.081254
29	Function	85	0.075897
30	Property	65	0.058039
31	Directory	50	0.044645
32	Branch	43	0.038395
33	FunctionParameter	40	0.035716
34	Parameter	33	0.029466
35	Package	29	0.025894
36	Local	28	0.025001
37	Variable	24	0.021430
38	jqAssistant	20	0.017858
39	Concept	19	0.016965

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 334426

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	87878	26.277263
1	MODIFIES	87878	26.277263
2	UPDATES	58386	17.458571
3	COMMITTED	22514	6.732132
4	CREATES	20734	6.199877
5	HAS_PARENT	12337	3.689007
6	DELETES	12112	3.621728
7	HAS_COMMIT	11257	3.366066
8	HAS_FILE	5891	1.761526
9	RENAMES	3354	1.002912
10	HAS_NEW_NAME	1770	0.529265
11	HAS_TAG	1732	0.517902
12	ON_COMMIT	1732	0.517902
13	HAS_AUTHOR	1265	0.378260
14	DEPENDS_ON	887	0.265231
15	HAS_KEY	668	0.199745
16	HAS_VALUE	668	0.199745
17	CONTAINS	596	0.178216
18	HAS_COMMITTER	370	0.110637
19	OF_TYPE	339	0.101368
20	EXPORTS	309	0.092397
21	REFERENCES	197	0.058907
22	DECLARES	186	0.055618
23	DECLARES_DEV_DEPENDENCY	169	0.050534
24	DECLARES_DEPENDENCY	161	0.048142
25	COPIES	138	0.041265
26	HAS_MEMBER	104	0.031098
27	HAS_TYPE_ARGUMENT	94	0.028108
28	DECLARES_SCRIPT	91	0.027211
29	RETURNS	82	0.024520

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

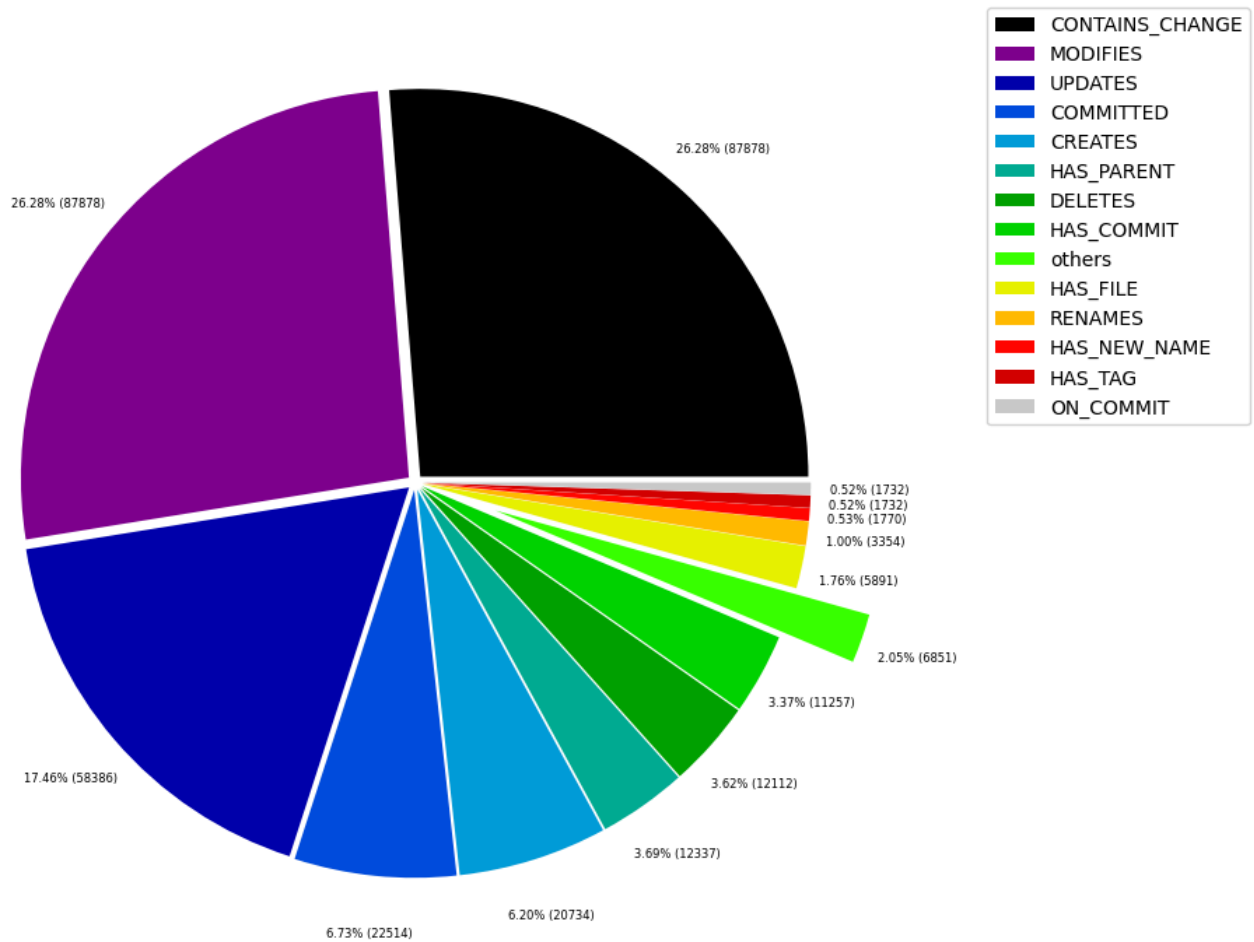


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

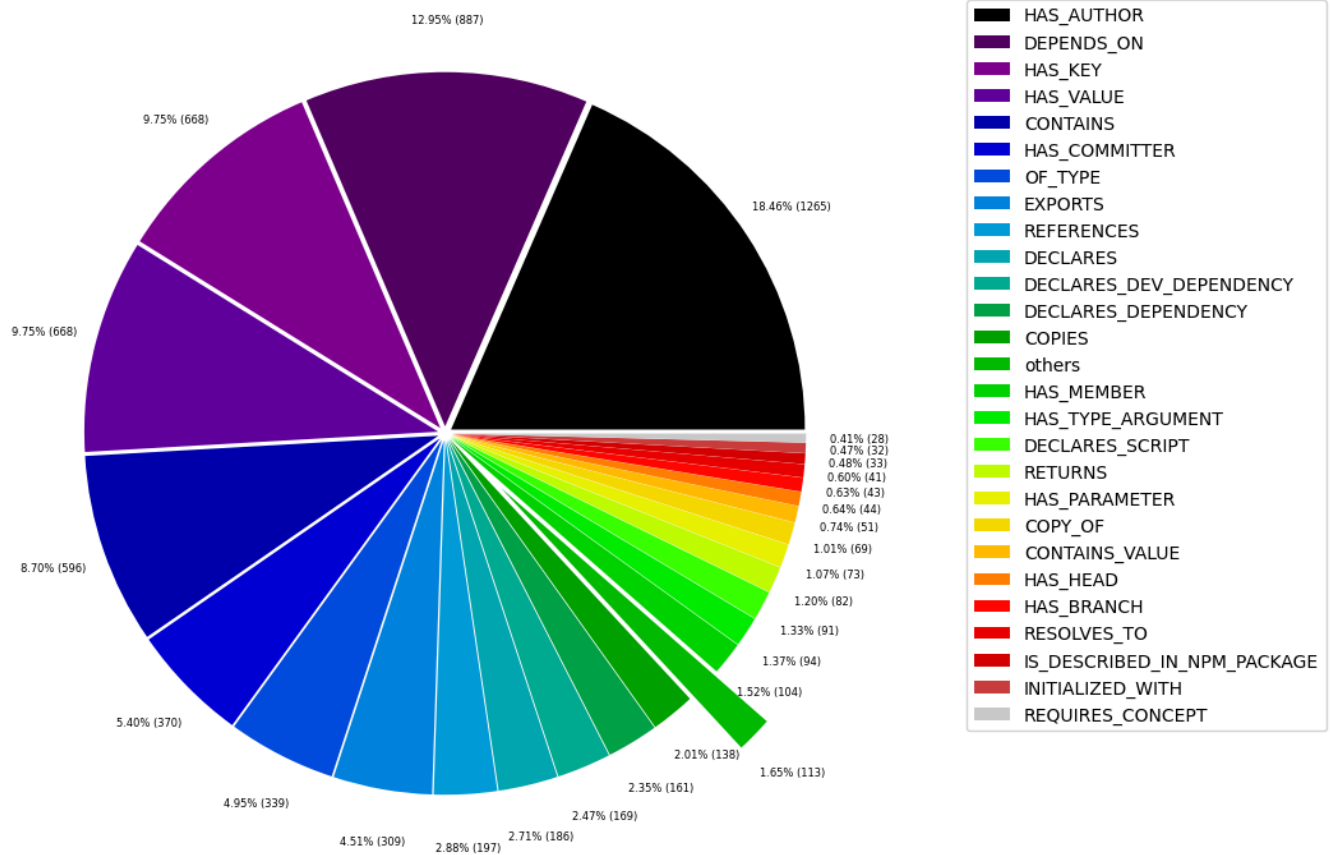
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000299
1	CONSTRAINED_BY	4	0.001196
2	IN_REPOSITORY	5	0.001495
3	REFERENCED_PROJECTS	5	0.001495
4	PARENT	6	0.001794
5	MEMBER	6	0.001794
6	HAS_ROOT	6	0.001794
7	HAS_NPM_PACKAGE	6	0.001794
8	HAS_CONFIG	6	0.001794
9	HAS_ARGUMENT	6	0.001794
10	EXTENDS	6	0.001794
11	DECLARES_ENGINE	6	0.001794
12	CONTAINS_PROJECT	6	0.001794
13	CALLS	6	0.001794
14	DECLARES_PEER_DEPENDENCY	8	0.002392
15	USES	11	0.003289
16	INCLUDES_CONCEPT	19	0.005681
17	REQUIRES_CONCEPT	28	0.008373
18	INITIALIZED_WITH	32	0.009569
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009868
20	RESOLVES_TO	41	0.012260
21	HAS_BRANCH	43	0.012858
22	HAS_HEAD	44	0.013157
23	CONTAINS_VALUE	51	0.015250
24	COPY_OF	69	0.020632
25	HAS_PARAMETER	73	0.021828
26	RETURNS	82	0.024520
27	DECLARES_SCRIPT	91	0.027211
28	HAS_TYPE_ARGUMENT	94	0.028108
29	HAS_MEMBER	104	0.031098

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	58386	11257	
1	[Git, Change, Update]	UPDATES	[File, Git]	58386	58386	
2	[Git, Change, Update]	MODIFIES	[File, Git]	58386	58386	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17242	11257	
4	[Git, Change, Create]	CREATES	[File, Git]	17242	17242	
5	[Git, Change, Create]	MODIFIES	[File, Git]	17242	17242	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12337	11257	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11257	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11257	1265	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11257	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8758	11257	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8758	8758	
12	[Git, Change, Delete]	DELETES	[File, Git]	8758	8758	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5891	1	
14	[Git, Change, Rename]	RENAMES	[File, Git]	3354	3354	
15	[Git, Change, Rename]	DELETES	[File, Git]	3354	3354	
16	[Git, Change, Rename]	CREATES	[File, Git]	3354	3354	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3354	3354	
18	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3354	11257	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5891	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1732	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1732	1732	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1265	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

```

total_number_of_nodes (vertices): 111994
total_number_of_relationships (edges): 334426
-> total directed graph density: 2.666333073100163e-05
-> total directed graph density in percent: 0.002666333073100163

```